MART-NILM: A Compact Multi-Appliance Representation Temporal Network for Joint Power-State Estimation in Non-Intrusive Load Monitoring

Raymond Siou Hin Tie, Wen-Shan Tan, Ze Yang Ding, Mirza Rayana Sanzana, and Haval Kamil

***Abstract*-Joint multi-appliance non-intrusive load monitoring (NILM) estimates appliance power and multi-label ON-OFF states from aggregate measurements using a single model. However, compact joint learning is challenged by sparse appliance activations observed under limited background-demand conditions and unequal power-state loss scales across appliances. This paper proposes the Multi-Appliance Representation Temporal Network for NILM (MART-NILM), combining shared temporal encoding, appliance-specific decoding, and cross-appliance feature exchange. Two training-only strategies address these challenges. Focal-event background-stratified mixing pairs selected ON-event traces with recorded residual backgrounds sampled across power strata, while bounded appliance-wise balancing calibrates power and state loss contributions separately for each appliance. Experiments on the UK Domestic Appliance-Level Electricity (UK-DALE) and REFIT datasets, with disjoint training, validation, and test houses, show that MART-NILM outperforms three reproduced joint multi-appliance baselines in appliance-average power error and macro-F1. It achieves mean absolute errors of 7.3 and 10.9 W and macro-F1 scores of 0.88 and 0.72, respectively. Compared with the strongest baseline, ON-state mean absolute error decreases by 58.4% and 34.4%, with 54.6% fewer parameters (1.373 million). These results support an improved accuracy-capacity trade-off for joint residential appliance monitoring.**

***Index Terms*-Energy disaggregation, multi-appliance estimation, multi-label classification, multi-task learning, non-intrusive load monitoring, power-state estimation.**

I. Introduction

Effective residential demand-side management requires appliance-level visibility into both electricity consumption and operating states. Power estimates quantify appliance demand, whereas state estimates expose activation timing; aggregate metering reveals neither directly. Appliance-level feedback can support energy conservation [1], but dedicated submetering requires installing and maintaining additional sensors. Non-intrusive load monitoring (NILM) estimates appliance-level power consumption and operating states from aggregate measurements without additional appliance-level sensors [2]. Joint estimation of these outputs is important because a low average power error can still mask missed operating periods, especially for sparsely active appliances.

Classical NILM identifies appliance signatures and infers appliance states using finite-state or factorial hidden Markov models [3], [4]. Deep learning approaches extract representations from aggregate measurements [5], reducing reliance on manually specified appliance signatures and state-transition models. Regardless of the method used, the aggregate power measured at the main meter includes demand from loads outside the monitored target set. For N target appliances, the aggregate power is expressed as

$$x_t = \sum_{i=1}^{N} P_{t,i} + b_t + \epsilon_t, \tag{1}$$

where $P_{t,i}$ is target power, b_t is background demand from unmonitored loads, and ϵ_t represents measurement and timing discrepancies. Background demand can exceed the power of low-consumption appliances and vary substantially across households, making appliance signatures difficult to distinguish from aggregate measurements. Consequently, an appliance activation observed under a particular background condition during training may appear under a different background condition in an unseen household.

Deep NILM approaches can be broadly categorized into single-appliance and multi-appliance estimation, with power regression and ON-OFF state detection as two complementary prediction tasks. Single-appliance approaches train separate predictors for individual loads [6], [7], duplicating temporal feature extraction and model storage as the number of target appliances increases. Multi-appliance approaches reduce this redundancy by jointly estimating the power traces and ON-state probabilities of multiple target appliances using a shared model [8]-[12]. However, sharing temporal representations introduces a trade-off between computational efficiency and appliance-specific feature learning.

Concurrent appliance activations provide useful contextual information, but overlapping power signatures and background demand can obscure individual targets [13]. An effective joint model must therefore preserve appliance-specific representations while exploiting shared temporal information and contextual interactions among appliances.

Beyond representation design, joint multi-appliance power-state learning faces two additional challenges related to

Raymond Siou Hin Tie, Wen-Shan Tan, and Ze Yang Ding are with the Department of Electrical and Robotics and the Centre for Net-Zero Technology, Monash University Malaysia, Bandar Sunway, Malaysia (e-mail: tan.wenshan@monash.edu).

Mirza Rayana Sanzana is with the School of Information Technology, Monash University Malaysia, Bandar Sunway, Malaysia.

Haval Kamil is with the Department of Air Conditioning and Refrigeration, College of Engineering Technology, Al-Kitab University, Kirkuk, Iraq.

training-data coverage and task optimization. First, sparse appliance activations may be observed under a limited range of background-demand conditions, restricting the event-background combinations available for training. Such limited coverage may encourage models to rely on incidental associations between appliance activations and background demand, which may not transfer to unseen households [13]. Second, appliances exhibit different power characteristics and activation frequencies, and their regression and classification losses can differ substantially in scale. Several existing joint NILM objectives employ fixed or globally adapted regression-classification weights [12], [14]. A shared task weight does not explicitly account for appliance-specific power-state loss ratios. When this weight is derived from aggregate losses, large regression errors from one appliance may influence the state-loss weighting applied to other appliances. These challenges motivate controlled event-background sampling and appliance-specific calibration of power-state learning objectives.

To address these challenges, we propose the Multi-Appliance Representation Temporal Network for NILM (MART-NILM), a compact framework for joint power estimation and multi-label ON-OFF state detection. MART-NILM integrates compact representation learning with two training-only strategies targeting event-background coverage and appliance-specific task imbalance.

The main contributions are threefold:

1) **Compact multi-appliance architecture:** MART-NILM employs a shared temporal encoder to avoid duplicating sequence modeling across target appliances. Lightweight appliance-specific decoders preserve specialized representations, while gated cross-appliance attention enables contextual feature exchange. The architecture jointly predicts five appliance power-state pairs in a single forward pass with 1.373 million parameters.
2) **Complementary training-only strategies:** Focal-event background-stratified mixing expands event-background coverage by pairing appliance traces sampled from windows containing ON activity with recorded residual backgrounds selected across power strata. Bounded appliance-wise balancing calibrates each appliance's state-classification loss relative to its power-regression loss, constraining the local loss ratio around a global reference. Neither strategy adds inference-time computation.
3) **Cross-house experimental validation:** MART-NILM is evaluated on two public residential datasets using disjoint training, validation, and test houses. Comparisons with three reproduced joint multi-appliance baselines, three-seed evaluations, and targeted ablation analyses assess prediction accuracy, training-strategy effectiveness, and computational efficiency. The results demonstrate an improved appliance-average power-state accuracy-capacity trade-off under the evaluated protocols.

II. Related Work

Viewed from a multi-task learning (MTL) perspective [15], joint multi-appliance power-state estimation combines two interacting design levels. Within each appliance, power regression and state classification must be coordinated; across appliances, feature sharing must reduce duplicated computation without suppressing appliance-specific information. This nested setting has been described as "multi-task-multi-appliance" [8] and "multi-appliance-task" [12], [16]. Prior NILM architectures are commonly distinguished by their temporal backbones, including convolutional neural networks (CNNs) [8], [16], recurrent neural networks (RNNs) [17], temporal convolutional networks (TCNs) [9], [18], and Transformers [12]. For joint multi-appliance power-state estimation, however, backbone choice addresses only temporal feature extraction. The central design questions are how the appliance head couples power regression with state classification and how the model body shares, specializes, and exchanges representations across appliances. Training strategies must additionally address sparse appliance events, limited background-demand coverage, and unequal power-state loss scales. The following review therefore considers appliance-head design, cross-appliance body design, and training strategies for joint power-state estimation.

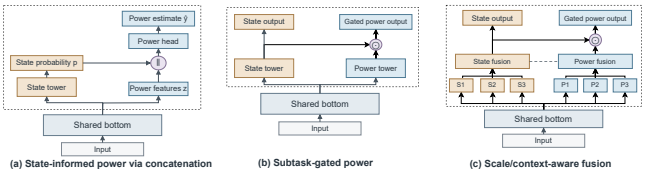


Fig. 1. Schematic within-appliance power-state heads: (a) state-informed concatenation (Murray *et al.* [17]); (b) output gating (SGN [19]); and (c) multi-scale fusion (SCANet [20]).

A. Appliance-Head Design

Existing joint power-state NILM studies have mainly examined how state information should influence power estimation within an individual target model. Figure 1 organizes representative appliance-head designs according to where this coupling is introduced.

Murray *et al.* [17] concatenate predicted states with power features (Fig. 1(a)), but the regressor remains free to produce nonzero power for an OFF prediction. The subtask-gated network (SGN) [19] instead gates regression outputs with ON-state probabilities (Fig. 1(b)), suppressing inactive power while making reconstruction sensitive to missed activations. The scale- and context-aware network (SCANet) [20] adds parallel dilated branches and self-attention (Fig. 1(c)), increasing processing within each target model. These designs address within-appliance power-state coupling, but their original single-appliance formulations require separate models for different targets, repeating aggregate feature extraction and providing no explicit exchange among appliance-specific representations.

B. Cross-Appliance Body Design

Cross-appliance body design determines which representations remain shared, which are decoded separately, and

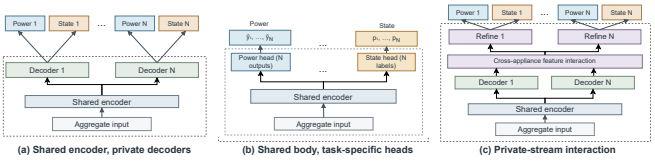


Fig. 2. Schematic multi-appliance body mechanisms: (a) shared encoder/private decoders (MATNilm [16] and Sun *et al.* [12]); (b) shared body/task heads (UNet-NILM [8]); and (c) private-stream interaction (MATNilm [16] and MART-NILM).

where appliance-specific streams exchange information. Multi-Appliance-Task NILM (MATNilm) [16] reduces this duplication through shared encoding and private appliance decoders (Fig. 2(a)). Its temporal and appliance-wise attention captures both dependencies, but repeats temporal attention and feed-forward processing in the private branches. Sun *et al.* [12] use the same sharing pattern with Transformer blocks in shared and appliance-specific layers (Fig. 2(a)). Downsampling reduces shared attention cost, but temporal attention is still repeated for individual appliances. Both support joint multi-appliance power and state estimation; their private sequence-processing blocks nevertheless add computation beyond the shared encoder.

UNet-NILM [8] follows the shared-body/task-head pattern (Fig. 2(b)), predicting multiple appliance powers and states from common features. It avoids private temporal blocks, but leaves appliance differentiation to shared features and output heads, without dedicated appliance-specific decoder streams or explicit exchange between such streams.

Cross-appliance interaction (Fig. 2(c)) complements these designs by exchanging context among specialized representations. MATNilm already uses appliance-wise attention for this purpose [16], while message passing provides a related mechanism in dense prediction [21]. These designs motivate examining how appliance-specific decoding and relation attention can be retained while reducing repeated temporal processing.

C. Training Strategies for Joint Power-State Estimation

Training a joint multi-appliance power-state model also requires sufficient exposure to appliance activations under different background-demand conditions. Neural NILM constructs synthetic aggregates to increase training diversity [5], SCANet augments ON-state examples [20], and aggregate recomposition varies background demand while preserving target labels [13]. These approaches address limited training coverage, but input diversity alone does not ensure that sparse ON events are sampled across a controlled range of recorded background demand. This distinction matters when event-background correlations change across houses.

A separate issue is the unequal scale of power and state losses. Gradient normalization (GradNorm) balances task updates in generic MTL [22], whereas SAMNet adapts a power-state loss ratio for joint NILM [14]. In a multi-appliance setting, a balance derived from aggregate losses may be driven by appliances with larger regression losses and may not reflect the power-state loss ratio of each appliance. This motivates appliance-wise balancing with bounds that limit extreme weights.

Together, these studies motivate the architectural and training choices examined in this paper. A shared temporal encoder, appliance-specific decoders, and relation attention address representation design, while focal-event mixing and bounded appliance-wise balancing address background-demand coverage and appliance-specific power-state loss scales, respectively.

III. Methodology

A. Problem Formulation

Following Equation (1), let $\mathbf{x} = (x_1, \ldots, x_T)$ denote an aggregate window of length T. The problem is to jointly estimate the power traces and ON-state probabilities of N target appliances:

$$\left(\hat{\mathbf{Y}}, \hat{\mathbf{S}}\right) = f_{\boldsymbol{\theta}}(\mathbf{x}), \qquad \hat{\mathbf{Y}} \in \mathbb{R}^{T\times N}, \quad \hat{\mathbf{S}} \in [0,1]^{T\times N} \tag{2}$$

Here $\boldsymbol{\theta}$ denotes the model parameters, and $\hat{\mathbf{Y}}$ and $\hat{\mathbf{S}}$ collect the estimated power traces and ON-state probabilities, respectively. To make the multi-appliance and multi-label output structure explicit, the time-t output slice of (2) is

$$f_{\boldsymbol{\theta},t}(\mathbf{x}) = \underbrace{\left(\hat{y}_{t,1}, \hat{s}_{t,1}, \ldots, \hat{y}_{t,N}, \hat{s}_{t,N}\right)}_{N \text{ appliance power-state pairs}}, \quad t = 1, \ldots, T. \tag{3}$$

Power estimation is a regression task, while state detection is formulated as multi-label classification, allowing multiple appliances to be active simultaneously. Binary ON-OFF labels describe appliance activity without distinguishing internal operating modes.

B. Model Architecture

The implementation uses instance-batch normalization (IBN), batch normalization (BN), the rectified linear unit (ReLU) activation, and fully connected (FC) output layers.

Figure 3 presents MART-NILM, combining the private-stream interaction body in Fig. 2(c) with the state-gated heads in Fig. 1(b). The components are described below in processing order, using the same names as the architecture-ablation variants in Table V.

Multi-scale CNN: Parallel convolutional branches [23] extract local features at complementary time scales from the standardized aggregate $\bar{\mathbf{x}}$:

$$\mathbf{Z} = \mathcal{F}([\mathcal{B}_3(\bar{\mathbf{x}}); \mathcal{B}_5(\bar{\mathbf{x}}); \mathcal{B}_9(\bar{\mathbf{x}})]) + \mathcal{S}(\bar{\mathbf{x}}). \tag{4}$$

Here $\mathcal{B}_k$ is a kernel-k convolution followed by IBN [24] and ReLU; $[\cdot;\cdot]$ denotes channel concatenation. The 1×1 fusion $\mathcal{F}$ combines the branches, while the projected shortcut $\mathcal{S}$ retains a direct input path. IBN applies instance normalization (IN) and BN to separate channel halves.

Feature Encoder: Two convolution-IBN-ReLU stages combine the local features into a shared representation $\mathbf{H}^{(0)} \in \mathbb{R}^{128\times T}$. Feature extraction is performed once for all appliances.

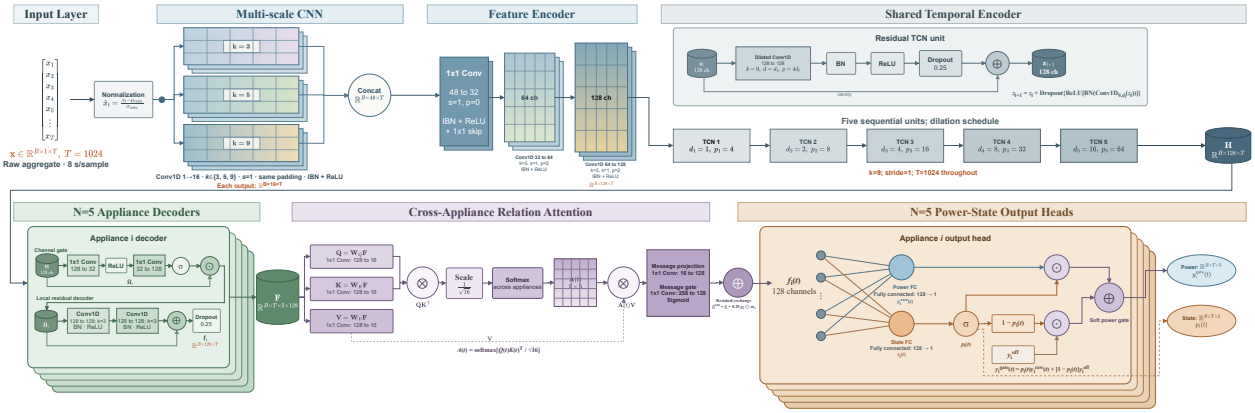


Fig. 3. Model architecture of MART-NILM.

Shared Temporal Encoder: Residual dilated TCN blocks [25] extend temporal context:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathcal{T}_{2^\ell}(\mathbf{H}^{(\ell)}), \qquad \ell \in \{0,1,2,3,4\}. \tag{5}$$

Here $\mathcal{T}_d$ comprises a dilation-d convolution, BN, ReLU, and dropout. Increasing dilation expands context, while residual connections and symmetric padding retain the incoming features and temporal alignment. The output is $\mathbf{H} = \mathbf{H}^{(5)}$.

Appliance Decoders: Each decoder applies a channel gate, inspired by channel recalibration [26], and a local residual transformation to obtain an appliance-specific representation $\mathbf{F}_i$. These private streams adapt the shared temporal features to individual load characteristics.

Cross-Appliance Relation Attention: The relation module draws on the cross-task information-sharing principle of the Multi-Tasks Guided Prediction-and-Distillation Network (PAD-Net) [21] to exchange context among appliance-specific feature streams. Scaled dot-product attention [27] is evaluated across appliances at each time t:

$$\mathbf{A}_t = \mathrm{softmax}\left(\frac{\mathbf{Q}_t\mathbf{K}_t^{\mathsf{T}}}{\sqrt{d}}\right), \qquad \mathbf{M}_t = \mathbf{A}_t\mathbf{V}_t. \tag{6}$$

Here $\mathbf{Q}_t$, $\mathbf{K}_t$, and $\mathbf{V}_t$ are learned projections of the stacked appliance representations, and d is their projection width. The attended context is projected back to the feature dimension and added to each private stream through a gated residual connection.

Power-State Output Heads: For each appliance, two time-shared FC heads predict raw standardized power $\hat{y}_i^{\mathrm{raw}}(t)$ and a state logit $a_i(t)$. Following the state-gating principle of SGN [19], the ON-state probability $\hat{s}_i(t) = \mathrm{sigmoid}(a_i(t))$ softly gates the power prediction:

$$\hat{y}_i(t) = \hat{s}_i(t)\hat{y}_i^{\mathrm{raw}}(t) + [1 - \hat{s}_i(t)]y_i^{(0)}, \tag{7}$$

where $y_i^{(0)} = -\mu_i/\sigma_i$ is the standardized zero-power anchor. Hence, a low ON-state probability smoothly moves the estimate towards zero power. Predictions are converted back to watts using the appliance training mean μ_i and standard deviation σ_i, and negative values are clipped to zero. The gate couples the state and power predictions while retaining a separate waveform regressor.

C. Loss Function

The model is trained jointly over all appliances, as in multi-task NILM [8], [12], [16]. Let $\mathcal{L}_{\mathrm{power},i}$ and $\mathcal{L}_{\mathrm{state},i}$ denote the power and state losses for appliance i, respectively. The joint objective is

$$\mathcal{L} = \sum_{i=1}^{N} \left(\mathcal{L}_{\mathrm{power},i} + \lambda r_i \mathcal{L}_{\mathrm{state},i}\right), \qquad \lambda = 0.8. \tag{8}$$

Here r_i is the appliance-wise balancing factor defined in Section III-D.

Power loss: The power objective combines waveform fidelity and switching fidelity:

$$\mathcal{L}_{\mathrm{power},i} = \underbrace{\mathcal{L}_{\mathrm{wave},i}}_{\text{power-level fidelity}} + \underbrace{0.15\,\mathcal{L}_{\mathrm{trans},i}}_{\text{transition fidelity}}. \tag{9}$$

The waveform and transition terms both use mean squared error (MSE), applied to power values and temporal differences, respectively. The expressions below are written for one window; in implementation, both terms are averaged over the batch and time dimensions. Let $y_i(t)$ and $\hat{y}_i(t)$ denote standardized target and predicted power, and let $z_i(t) \in \{0,1\}$ denote the ON-OFF label. For a temporal index set $\mathcal{S}$, define

$$\mathrm{MSE}_{\mathcal{S}}(u,v) = \frac{1}{\max(|\mathcal{S}|,1)} \sum_{t\in\mathcal{S}} \big(u(t) - v(t)\big)^2. \tag{10}$$

The waveform term evaluates power errors over all samples $\mathcal{T}_0 = \{1,\ldots,T\}$, ON-state samples $\mathcal{T}_i^{\mathrm{on}} = \{t \in \mathcal{T}_0 : z_i(t) = 1\}$, and OFF-state samples $\mathcal{T}_i^{\mathrm{off}} = \{t \in \mathcal{T}_0 : z_i(t) = 0\}$:

$$\begin{aligned}\mathcal{L}_{\mathrm{wave},i} = {} & \mathrm{MSE}_{\mathcal{T}_0}(y_i,\hat{y}_i) + \mathrm{MSE}_{\mathcal{T}_i^{\mathrm{on}}}(y_i,\hat{y}_i) \\ & + 0.5\ \mathrm{MSE}_{\mathcal{T}_i^{\mathrm{off}}}(y_i,\hat{y}_i).\end{aligned} \tag{11}$$

The first term measures global waveform fidelity; the ON-state term prevents sparse activations from being dominated by OFF samples, and the lower-weight OFF-state term constrains inactive predictions.

To additionally constrain changes in power, let $\Delta y_i(t) = y_i(t) - y_i(t-1)$ and define $\Delta\hat{y}_i(t)$ analogously. The transition term compares these differences whenever the appliance is ON at either adjacent sample, over $\mathcal{T}_i^{\Delta} = \{t \in \{2,\ldots,T\} : z_i(t) = 1 \text{ or } z_i(t-1) = 1\}$:

$$\mathcal{L}_{\mathrm{trans},i} = \mathrm{MSE}_{\mathcal{T}_i^{\Delta}}(\Delta y_i, \Delta\hat{y}_i). \tag{12}$$

State loss: The state objective combines weighted binary cross-entropy (BCE) with explicit false-positive control:

$$\mathcal{L}_{\text{state},i} = \underbrace{\mathcal{L}_{\text{BCE},i}}_{\text{class-balanced state detection}} + \underbrace{\mathcal{L}_{\text{fp},i}}_{\text{false-positive control}}. \tag{13}$$

For a mini-batch of B windows, let $\hat{s}_{b,t,i} \in (0,1)$ be the predicted ON-state probability and $z_{b,t,i} \in \{0,1\}$ the target label. The class-balanced BCE term uses $w_i = \min\{(1-\pi_i)/\pi_i, 12\}$, determined by the training ON-state proportion π_i:

$$\mathcal{L}_{\text{BCE},i} = -\frac{1}{BT}\sum_{b,t}\big[w_i z_{b,t,i}\log \hat{s}_{b,t,i} + (1-z_{b,t,i})\log(1-\hat{s}_{b,t,i})\big]. \tag{14}$$

The false-positive term penalizes confident ON-state predictions on OFF samples:

$$\mathcal{L}_{\text{fp},i} = \frac{\sum_{b,t}(1-z_{b,t,i})\hat{s}^2_{b,t,i}}{\max\left(\sum_{b,t}(1-z_{b,t,i}), 1\right)}. \tag{15}$$

D. Bounded Appliance-Wise Balancing

To account for the different numerical scales of regression and classification losses, a bounded loss-ratio weighting is used. It is motivated by loss-ratio balancing in SAMNet [14] and differs from gradient-norm balancing in GradNorm [22]. The global reference ratio g bounds the appliance-specific ratio r_i:

$$g = \frac{\sum_j \mathcal{L}_{\text{power},j}}{\sum_j \mathcal{L}_{\text{state},j}}, \qquad r_i = \min\left\{3g, \max\left\{\frac{g}{3}, \frac{\mathcal{L}_{\text{power},i}}{\mathcal{L}_{\text{state},i}}\right\}\right\}. \tag{16}$$

At each mini-batch update, g and r_i are evaluated from the current losses and held fixed during back-propagation. The three-fold bounds limit each local ratio to $[g/3, 3g]$, preventing extreme weights from dominating optimization.

E. Focal-Event Background-Stratified Mixing

Training uses recorded and recomposed aggregate windows with equal probability. Recorded windows preserve the observed joint demand. These windows extend synthetic aggregate construction in Neural NILM [5] by pairing appliance events with recorded background demand from other training windows [13], reducing reliance on house-specific event-background correlations. For a recorded sequence, the background demand is estimated by the nonnegative residual

$$R_t = \left[x_t - \sum_{i=1}^{N} P_{t,i}\right]_+. \tag{17}$$

Here $[u]_+ = \max(u, 0)$. Unlike b_t in Equation (1), R_t can also include measurement and timing discrepancies. For each recomposed window, the residual and appliance traces are drawn from independently selected training windows:

$$\tilde{x}_t = R_t^{(k_0)} + \sum_{i=1}^{N} P_{t,i}^{(k_i)}, \qquad \tilde{P}_{t,i} = P_{t,i}^{(k_i)}, \quad \tilde{z}_{t,i} = z_{t,i}^{(k_i)}, \tag{18}$$

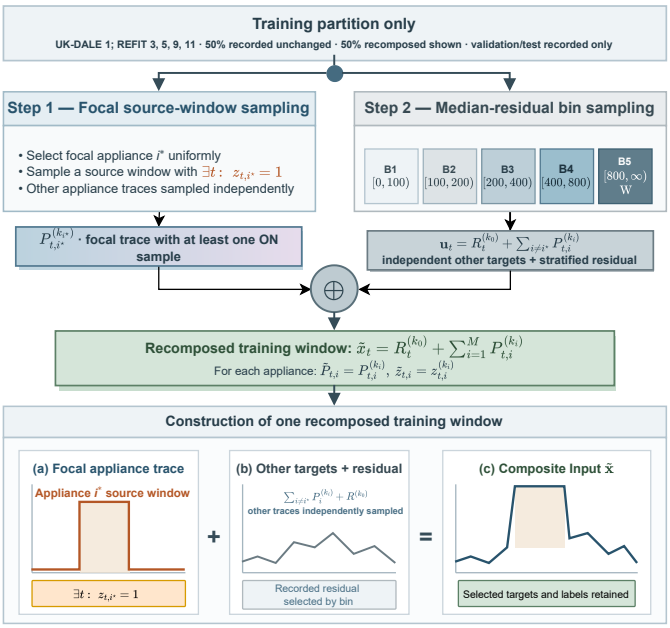


Fig. 4. Focal-event background-stratified mixing. A focal ON-event trace, independently sampled target traces, and a residual window selected by median-power bin form a recomposed training window.

where $k_0, k_1, \ldots, k_N$ denote independently sampled window indices. In Step 1 of Fig. 4, one focal appliance $i^\star$ is selected uniformly, and its trace is drawn from a window containing at least one ON sample; the remaining appliance traces are sampled independently. In Step 2, a non-empty bin of median residual power is selected uniformly from $[0, 100)$, $[100, 200)$, $[200, 400)$, $[400, 800)$, and $[800, \infty)$ W, and a background window is sampled uniformly within that bin. The selected traces and background are then added according to Equation (18), as illustrated in the lower panel of Fig. 4. The power and state labels of every selected appliance trace are retained. Recomposition is used only during training; validation and test use recorded aggregates and aligned labels.

IV. Experimental Evaluation

A. Dataset

Two public residential datasets are used: the UK Domestic Appliance-Level Electricity (UK-DALE) dataset [28] and the REFIT electrical-load dataset [29]. The target appliances are the kettle, fridge, dishwasher, washer, and microwave, and all signals are resampled at 8 s.

The experiment uses a whole-house split: UK-DALE house 1 and REFIT houses 3, 5, 9, and 11 for training; UK-DALE house 5 and REFIT house 2 for validation; and UK-DALE house 2 and REFIT house 20 for testing. The resulting training-validation-test ratio is 5:2:2 by house count. Because each house contributes one continuous 5-week block, this also corresponds to 55.6%, 22.2%, and 22.2% of the nominal recording duration, respectively. The held-out blocks are centred within the previously established eight-week recording intervals and are selected without reference to appliance events or model performance. Normalization statistics are estimated from the training houses only.

TABLE I
ON-OFF LABEL PARAMETERS. τ^P, G, AND D DENOTE THE ON-POWER THRESHOLD, MAXIMUM OFF-GAP DURATION, AND MINIMUM ON DURATION, RESPECTIVELY. THE FINAL COLUMN LISTS UK-DALE / REFIT VALUES.

Appliance	τ^P (W)	G (s)	D (s)
Kettle	200	0	12 / 12
Fridge†	50	12	60 / 60
Dishwasher†	50	1800	960 / 960
Washer†	20	960	960 / 960
Microwave	200	30	12 / 6

† For each sample, a five-sample local-median filter sets its label-generation power to zero when at least 60% of its four surrounding values are below 50 W and the sample exceeds both three times their median and 100 W. Endpoint windows use edge replication.

B. Data Preprocessing

After 8 s resampling, no additional temporal alignment is applied; residual timing discrepancies remain a measurement limitation. Rows missing any required aggregate, appliance-power, or state-label value are excluded, and windows do not cross timestamp discontinuities.

Let n index windows and t index samples within a window. The aggregate mean and standard deviation, μ_x and σ_x, and the corresponding appliance-i statistics, μ_i and σ_i, are estimated from the training houses. The standardized input power, $\bar{x}_{n,t}$, and target power, $y_{n,t,i}$, are defined as

$$\bar{x}_{n,t} = \frac{x_{n,t} - \mu_x}{\sigma_x}, \qquad y_{n,t,i} = \frac{P_{n,t,i} - \mu_i}{\sigma_i}. \tag{19}$$

State labels are precomputed before windowing. For appliances marked in Table I, isolated spikes are first removed on a label-only copy of the resampled power trace. Each sample is then labeled ON when this trace reaches its appliance-specific threshold τ_i^P. OFF gaps of at most G_i are merged; subsequently, complete ON runs shorter than D_i are removed. Runs touching a sequence boundary are retained because their full duration is unobserved. Table I specifies the physical-time values of these parameters, which are converted to 8 s samples by half-up rounding before labeling. UK-DALE house 5 uses a 30 s fridge gap.

C. Experimental Setup

All experiments were conducted on a single NVIDIA GeForce RTX 4090 graphics processing unit (GPU; 24 GB) using PyTorch 2.6.0 with Compute Unified Device Architecture (CUDA) 12.4. MART-NILM is compared with UNet-NILM [8], the convolutional MATNilm variant (MAT-Conv) [16], and the multi-appliance-task network of Sun *et al.* [12]. Each baseline retains the architecture, objective, and training hyperparameters of its source paper, while all models use the common cross-house data protocol. Table II lists their settings.

MART-NILM is optimized with Adam using an initial learning rate of 10^{-4}, weight decay of 10^{-4}, and unit-norm gradient clipping. The learning rate is reduced by a factor of 0.5 after 12 epochs without improvement in the validation score, to a minimum of 10^{-5}. At 8 s per sample, each 1024-sample input-output window spans 136.5 min of household measurements.

TABLE II
TRAINING WINDOWING AND BATCH SETTINGS. ALL WINDOW LENGTHS AND STRIDES ARE IN SAMPLES.

Model	Input	Output	Stride	Batch	Optimizer	Epochs
UNet-NILM [8]	100	1	1	128	Adam	50
MAT-Conv [16]	864	64	64	64	Adam	200
Sun *et al.* [12]	480	480	240	32	Adam	100
MART-NILM	1024	1024	512	64	Adam	150

For MART-NILM, checkpoint selection uses only the validation houses. Mean absolute error (MAE) in standardized power coordinates and average precision (AP) are computed from the validation outputs, and the retained checkpoint minimizes

$$S_{\text{val}} = \text{MAE}_{\text{val}}^{\text{norm}} + 1 - \text{AP}_{\text{val}}^{\text{macro}}, \tag{20}$$

where $\text{MAE}_{\text{val}}^{\text{norm}}$ is the macro mean absolute error over appliances in standardized power coordinates, and the AP term summarizes the precision-recall curve without selecting a decision threshold. For each model and appliance i, let $\hat{s}_{t,i} \in [0,1]$ denote the predicted ON probability at sample t. After checkpoint selection, an appliance-specific threshold $\gamma_i \in \{0.05, 0.06, \ldots, 0.98\}$ is selected on the validation houses to maximize appliance-level F1 after fixed temporal post-processing is applied. At test time, the decoded state label, $\hat{z}_{t,i}$, is obtained as

$$\hat{z}_{t,i} = \mathbb{I}[\hat{s}_{t,i} \geq \gamma_i], \tag{21}$$

where $\mathbb{I}[\cdot]$ is the indicator function. For the microwave, fixed temporal post-processing removes predicted ON runs shorter than 32 s and merges OFF gaps of at most 24 s. The decoded state is also applied to the power output, $\hat{P}_{t,i}^{\text{eval}} = \hat{z}_{t,i}\hat{P}_{t,i}$. Thresholds and temporal rules are frozen before testing and applied unchanged to both test houses; test labels are not used to select γ_i.

D. Evaluation Metrics

Each appliance is evaluated on power and on the ON-OFF state. For an evaluated record of length T_{rec}, P_t is the recorded power in watts and $\hat{P}_t^{\text{eval}}$ is the state-decoded prediction. The label z_t equals 1 when the appliance is recorded ON and 0 when it is OFF. $\hat{z}_t$ is the predicted label.

Metrics are computed separately for the held-out UK-DALE house 2 and REFIT house 20. Within each test house, the average (Avg.) column is the macro average over the five appliances; architecture-ablation summaries give the two houses equal weight. Performance metrics are reported as mean $\pm$ standard deviation over three random seeds on the fixed cross-house split, so the standard deviation reflects optimization variability. The evaluation considers complementary power-estimation and state-detection metrics.

1) Power: Power error is reported using MAE, ON-state MAE (ON MAE), and blockwise signal aggregate error (SAE). MAE uses every sample; ON MAE is the same quantity evaluated only at samples with $z_t = 1$, so that reconstruction quality during appliance operation remains visible:

$$\mathrm{MAE} = \frac{1}{T_{\mathrm{rec}}} \sum_{t=1}^{T_{\mathrm{rec}}} \left| P_t - \hat{P}_t^{\mathrm{eval}} \right|. \tag{22}$$

Following MATNilm [16], SAE measures the mean absolute error of average power over $K = \lfloor T_{\mathrm{rec}}/L \rfloor$ non-overlapping 160 min blocks ($L = 1200$ samples):

$$\mathrm{SAE} = \frac{1}{K} \sum_{k=1}^{K} \left| \frac{1}{L} \sum_{t=(k-1)L+1}^{kL} \left(P_t - \hat{P}_t^{\mathrm{eval}} \right) \right|, \tag{23}$$

2) On-Off State: The recorded label z_t follows the rules in Table I, and $\hat{z}_t$ is the predicted binary state. For each appliance, TP (true positives) counts samples correctly identified as ON ($z_t = 1, \hat{z}_t = 1$), FP (false positives) counts inactive samples falsely detected as ON ($z_t = 0, \hat{z}_t = 1$), and FN (false negatives) counts active samples missed as OFF ($z_t = 1, \hat{z}_t = 0$):

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{24}$$

$$\mathrm{F1} = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{25}$$

Precision is the fraction of predicted ON samples that are truly ON, and recall is the fraction of actual ON samples detected. F1 is their harmonic mean; true negatives (TN, $z_t = 0, \hat{z}_t = 0$) are excluded to prevent long idle background periods from artificially inflating the score. To evaluate state ranking independently of the calibrated decision threshold, we also report AP from the continuous state probability. Let $\mathcal{P}_n$ and $\mathcal{R}_n$ denote precision and recall after the nth distinct probability threshold, with $\mathcal{R}_0 = 0$. The non-interpolated AP is

$$\mathrm{AP} = \sum_{n} (\mathcal{R}_n - \mathcal{R}_{n-1})\, \mathcal{P}_n. \tag{26}$$

Unlike F1, AP does not use the appliance-specific threshold or the temporal state post-processing fitted on the validation houses.

V. Results and Discussion

A. Joint Power and State Estimation

Table III compares joint power-estimation and ON-OFF state-detection performance on the two held-out test houses. A displayed zero standard deviation indicates a value below the shown precision.

MART-NILM achieves the lowest appliance-average MAE and highest macro-F1 on both test houses among the evaluated models, outperforming MAT-Conv, the strongest baseline in overall performance. Relative to MAT-Conv, MART-NILM reduces overall MAE by 23.6% and 14.5% on UK-DALE and REFIT, respectively. More pronounced improvements are observed during appliance operation, with ON MAE decreasing by 58.4% and 34.4%, respectively. Blockwise SAE also decreases by 27.0% and 27.8%. State-detection performance improves consistently at the macro level: macro-F1 increases from 0.72 to 0.88 on UK-DALE and from 0.58 to 0.72 on REFIT. Threshold-independent macro-AP also rises from 0.812 to 0.926 and from 0.576 to 0.761, respectively, indicating improvements in state-probability ranking beyond the selected decision thresholds. MART-NILM also records the highest mean AP across all ten appliance-dataset combinations, although the margins are small for some appliances. Together with the macro-F1 improvements, these results indicate that the state-detection gains extend beyond threshold selection.

The improvements are not uniform across individual appliances. MAT-Conv retains lower MAE for the UK-DALE fridge and REFIT kettle and microwave, while its REFIT fridge F1 is marginally higher. Nevertheless, the REFIT microwave illustrates the importance of joint power-state evaluation: although MART-NILM has a higher overall MAE (6.8 versus 6.0 W), it substantially reduces ON MAE from 1062.2 to 429.0 W and improves F1 from 0.23 to 0.56. These appliance-level differences show that the complete framework improves the overall balance between power estimation and state detection, rather than uniformly outperforming the baselines on every metric.

Figure 5 provides a qualitative view of the same comparison. MART-NILM generally follows activation boundaries and multi-stage plateaus while suppressing inactive-period power, although amplitude and timing errors remain.

B. Performance and Model Efficiency

Table IV shows that MART-NILM uses only 1.373 million parameters, representing reductions of 54.6%, 76.5%, and 82.5% relative to MAT-Conv, UNet-NILM, and Sun *et al.*, respectively. Together with the improvements in appliance-average MAE and macro-F1 in Table III, these results demonstrate a favorable accuracy-capacity trade-off: stronger joint prediction performance is achieved with substantially fewer model parameters. MART-NILM also has the smallest checkpoint and the shortest observed training and end-to-end inference times under the evaluated protocols. The complete prediction pipeline processes both test records in 1.14 s, compared with 14.67 s for MAT-Conv, 23.50 s for Sun *et al.*, and 80.40 s for UNet-NILM. The benchmark includes data transfer and output reconstruction under each model's native windowing protocol, so it measures the complete prediction pipeline rather than isolated forward-pass latency. MART-NILM benefits from producing long output sequences: its 1024-sample windows advance by 512 samples, requiring substantially fewer windows than sequence-to-point inference while preserving dense temporal predictions.

C. Architecture Ablation Study

Each variant in Table V removes or simplifies one top-level component while preserving the joint five-appliance output structure and compatible feature dimensions.

The components make complementary contributions rather than improving every metric uniformly. Temporal modeling has the largest effect: removing it increases MAE by 1.44 W

TABLE III

POWER AND ON-OFF RESULTS (MEAN ± STANDARD DEVIATION OVER THREE SEEDS; MAE, ON MAE, AND SAE ARE IN W). BOLD DENOTES THE BEST DISPLAYED MEAN, INCLUDING TIES.

Metric	Method	UK-DALE						REFIT					
		Kettle	Fridge	Dishwasher	Washer	Microwave	Avg.	Kettle	Fridge	Dishwasher	Washer	Microwave	Avg.
MAE	UNet-NILM	9.2±2.1	29.1±0.9	22.6±5.1	9.1±0.1	6.1±0.3	15.2±1.1	8.6±0.9	35.0±1.7	12.1±1.1	7.6±0.1	7.4±0.3	14.2±0.3
	MAT-Conv	5.5±1.6	**18.4±0.7**	8.9±2.1	9.6±1.4	5.3±0.4	9.5±1.1	**5.4±0.4**	40.8±0.1	5.4±0.8	6.2±2.3	**6.0±0.7**	12.8±0.6
	Sun *et al.*	8.8±1.7	34.4±3.0	14.7±6.2	15.5±5.0	6.3±0.1	15.9±1.3	10.9±1.6	35.6±1.8	9.0±1.2	18.0±7.3	6.7±0.1	16.1±0.8
	MART-NILM	**3.8±0.2**	19.8±0.7	**4.2±0.4**	**5.3±0.3**	**3.4±0.1**	**7.3±0.2**	5.9±0.1	**33.0±1.2**	**4.9±0.2**	**4.1±0.8**	6.8±0.1	**10.9±0.2**
SAE	UNet-NILM	8.4±2.6	24.5±2.5	21.1±6.5	9.0±0.1	6.0±0.4	13.8±1.3	5.6±1.1	25.9±2.6	9.8±0.3	7.1±0.3	7.0±0.2	11.1±0.7
	MAT-Conv	4.3±1.2	**10.0±1.0**	6.7±1.4	8.8±1.3	5.0±0.4	6.9±0.8	2.9±0.3	23.7±0.6	4.5±1.0	5.0±1.8	**5.2±0.5**	8.3±0.5
	Sun *et al.*	7.3±1.4	32.1±3.5	12.0±5.9	14.5±4.9	6.2±0.1	14.4±1.4	7.1±1.1	21.7±4.4	7.9±1.3	13.0±5.5	6.6±0.0	11.3±0.7
	MART-NILM	**2.9±0.1**	12.8±1.5	**2.6±0.7**	**4.0±0.4**	**2.9±0.1**	**5.1±0.4**	**2.6±0.1**	**15.4±1.6**	**3.7±0.2**	**3.1±0.9**	5.2±0.1	**6.0±0.2**
ON MAE	UNet-NILM	1296.5±443.8	48.4±4.0	747.7±180.9	462.6±7.3	1178.8±93.3	746.8±110.0	1289.1±190.7	63.4±3.4	398.3±93.1	306.1±6.7	1196.7±58.0	650.7±59.1
	MAT-Conv	547.5±180.2	**22.1±1.6**	206.9±46.2	385.7±59.1	946.4±99.8	421.7±75.2	850.7±44.4	46.3±2.5	208.2±21.0	195.6±10.1	1062.2±90.6	472.6±29.3
	Sun *et al.*	807.3±69.1	61.0±7.1	449.8±212.8	478.2±28.0	1215.2±19.2	602.3±44.3	1012.0±65.7	59.1±6.4	437.1±91.3	306.7±17.6	1224.2±5.6	607.8±16.5
	MART-NILM	**383.4±15.9**	25.4±2.1	**87.0±7.4**	**161.8±17.0**	**218.8±23.9**	**175.3±6.9**	**785.2±17.4**	**40.4±1.5**	**139.5±6.0**	**155.9±5.3**	**429.0±40.8**	**310.0±3.5**
AP	UNet-NILM	0.908±0.016	0.873±0.041	0.580±0.274	0.866±0.026	0.535±0.024	0.753±0.049	0.653±0.036	0.657±0.030	0.269±0.056	0.654±0.129	0.136±0.046	0.474±0.024
	MAT-Conv	0.968±0.020	0.951±0.006	0.868±0.039	0.555±0.191	**0.718±0.063**	0.812±0.051	0.801±0.018	0.537±0.005	0.736±0.040	0.653±0.066	0.152±0.046	0.576±0.024
	Sun *et al.*	0.930±0.016	0.915±0.030	0.838±0.094	0.207±0.128	0.514±0.194	0.681±0.075	0.660±0.090	0.641±0.053	0.615±0.040	0.123±0.019	0.160±0.069	0.440±0.000
	MART-NILM	**0.997±0.000**	**0.970±0.001**	**0.959±0.002**	**0.985±0.007**	**0.722±0.046**	**0.926±0.010**	**0.848±0.007**	**0.736±0.030**	**0.759±0.014**	**0.914±0.010**	**0.547±0.056**	**0.761±0.016**
Precision	UNet-NILM	0.80±0.09	0.74±0.09	0.55±0.19	0.84±0.09	0.57±0.07	0.70±0.03	0.63±0.08	0.60±0.12	0.26±0.06	**0.89±0.05**	0.18±0.07	0.51±0.04
	MAT-Conv	0.82±0.10	0.76±0.03	0.47±0.05	0.61±0.18	**0.71±0.09**	0.67±0.08	0.70±0.10	0.59±0.00	0.42±0.06	0.59±0.07	0.16±0.03	0.49±0.04
	Sun *et al.*	0.74±0.08	0.87±0.01	0.51±0.16	0.22±0.09	0.55±0.49	0.58±0.11	0.57±0.08	0.63±0.00	0.54±0.09	0.14±0.02	0.37±0.28	0.45±0.04
	MART-NILM	**0.93±0.01**	**0.89±0.02**	**0.65±0.03**	**0.92±0.01**	**0.72±0.00**	**0.82±0.00**	**0.74±0.01**	**0.64±0.01**	**0.63±0.02**	0.83±0.07	**0.46±0.02**	**0.66±0.01**
Recall	UNet-NILM	0.87±0.08	0.82±0.14	0.65±0.21	0.78±0.04	0.57±0.07	0.74±0.05	0.73±0.03	0.65±0.18	0.46±0.07	0.43±0.13	0.37±0.04	0.53±0.07
	MAT-Conv	0.97±0.01	**0.98±0.02**	0.98±0.02	0.49±0.18	0.67±0.13	0.82±0.02	0.78±0.03	**0.93±0.06**	**0.83±0.02**	0.76±0.21	0.42±0.25	0.74±0.11
	Sun *et al.*	0.92±0.03	0.94±0.01	0.89±0.14	0.56±0.06	0.08±0.07	0.68±0.05	0.75±0.05	0.78±0.01	0.77±0.08	0.50±0.12	0.03±0.02	0.57±0.01
	MART-NILM	**1.00±0.00**	0.94±0.01	**0.99±0.01**	**0.97±0.01**	**0.90±0.02**	**0.96±0.00**	**0.82±0.01**	0.78±0.04	0.75±0.03	**0.88±0.03**	**0.72±0.04**	**0.79±0.02**
F1	UNet-NILM	0.83±0.03	0.77±0.07	0.60±0.20	0.81±0.03	0.57±0.04	0.71±0.03	0.67±0.03	0.61±0.08	0.33±0.07	0.58±0.12	0.23±0.05	0.48±0.04
	MAT-Conv	0.89±0.06	0.86±0.01	0.63±0.04	0.54±0.17	0.68±0.06	0.72±0.05	0.73±0.05	**0.72±0.02**	0.55±0.05	0.65±0.06	0.23±0.08	0.58±0.02
	Sun *et al.*	0.82±0.04	**0.91±0.00**	0.64±0.14	0.30±0.09	0.15±0.13	0.56±0.05	0.64±0.03	0.70±0.00	0.63±0.05	0.21±0.01	0.06±0.04	0.45±0.00
	MART-NILM	**0.96±0.00**	**0.91±0.00**	**0.79±0.02**	**0.94±0.01**	**0.80±0.01**	**0.88±0.00**	**0.78±0.01**	0.71±0.01	**0.69±0.00**	**0.85±0.03**	**0.56±0.03**	**0.72±0.00**

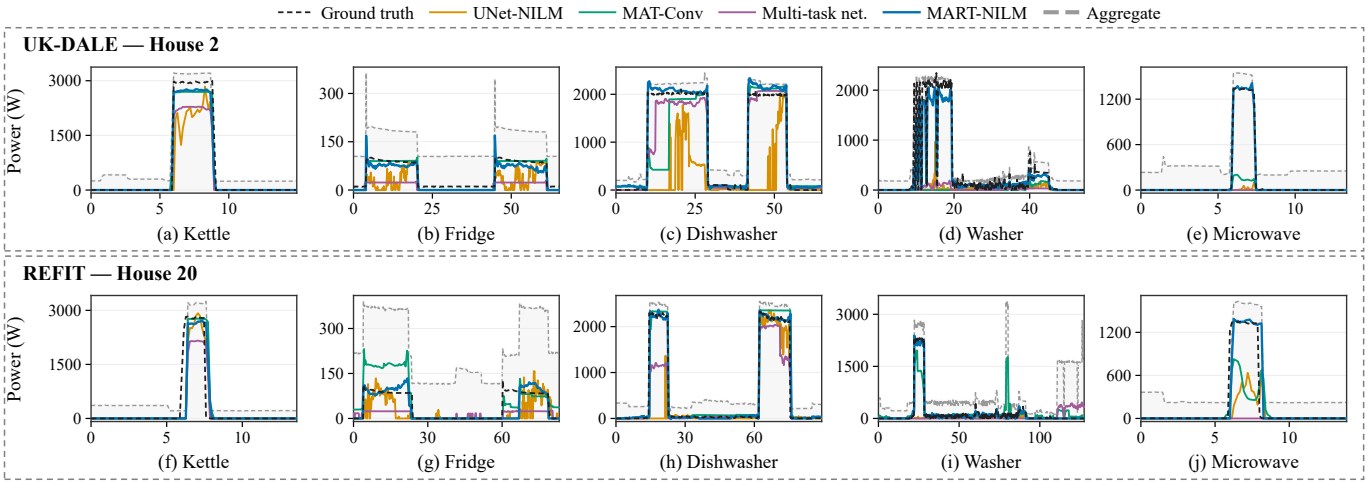


Fig. 5. Waveform comparisons on the two test houses.

TABLE IV

MODEL SIZE AND OBSERVED COMPUTATIONAL COST. M AND MB DENOTE MILLIONS OF PARAMETERS AND MEGABYTES, RESPECTIVELY.

Model	Params (M)	Size (MB)	Training (s)	Inference (s)
UNet-NILM	5.832	22.299	17,455.8	80.40
MAT-Conv	3.022	11.611	10,354.9	14.67
Sun *et al.*	7.829	59.793	16,493.8	23.50
MART-NILM	**1.373**	**5.335**	**485.2**	**1.14**

TABLE V

ARCHITECTURE ABLATION (THREE-SEED MEAN ± STANDARD DEVIATION OVER THE TWO TEST HOUSES; MAE, ON MAE, AND SAE ARE IN W).

Variant	Params (M)	MAE ↓	ON MAE ↓	SAE ↓	Macro-F1 ↑
Single-scale CNN	1.372	9.22 ±0.18	241.75 ±8.76	5.66 ±0.17	0.795 ±0.001
Without feature encoder	1.326	9.16 ±0.06	252.01 ±2.21	5.59 ±0.07	0.790 ±0.006
Without shared temporal encoder	0.634	10.56 ±0.33	288.07 ±10.25	6.56 ±0.16	0.710 ±0.013
Without appliance decoders	0.836	9.22 ±0.66	265.84 ±4.30	5.63 ±0.63	0.783 ±0.015
Without cross-appliance relation attention	1.332	9.18 ±0.09	**230.84 ±14.69**	**5.55 ±0.04**	0.790 ±0.015
MART-NILM	1.373	**9.12 ±0.14**	242.94 ±5.35	5.58 ±0.17	**0.796 ±0.008**

and lowers macro-F1 from 0.796 to 0.710. Removing the appliance-specific decoders increases ON MAE from 242.94 to 265.84 W and reduces macro-F1 from 0.796 to 0.783. These changes support retaining lightweight appliance-specific refinement alongside the shared temporal encoder. The feature encoder improves all four aggregate measures. Replacing the multi-scale CNN with a single-scale variant slightly worsens MAE, SAE, and macro-F1 while marginally improving ON MAE, indicating a modest rather than uniform contribution.

Cross-appliance relation attention provides a modest improvement in macro-F1 rather than a uniform reduction in

TABLE VI
TRAINING-STRATEGY ABLATION ON THE VALIDATION HOUSES (THREE-SEED MEAN ± STANDARD DEVIATION; MAE, ON MAE, AND SAE ARE IN W). † DENOTES A STRATEGY PROPOSED IN THIS WORK.

Training strategy	MAE ↓	ON MAE ↓	SAE ↓	Macro-F1 ↑
Standard joint training	21.164 ±0.394	357.808 ±19.280	15.938 ±0.348	0.605 ±0.012
+ Focal-event mixing†	21.679 ±0.144	**301.623 ±16.196**	15.601 ±0.365	0.649 ±0.016
+ Appliance-wise balancing†	20.929 ±0.065	334.862 ±10.888	15.175 ±0.322	0.661 ±0.014
Complete MART-NILM†	**20.791 ±0.175**	315.408 ±10.405	**15.016 ±0.385**	**0.687 ±0.003**

power errors. Removing the module lowers macro-F1 from 0.796 to 0.790 but slightly improves ON MAE and SAE, indicating a trade-off between state detection and power reconstruction. It also increases the across-seed standard deviation from 5.35 to 14.69 W for ON MAE and from 0.008 to 0.015 for macro-F1. Thus, cross-appliance exchange does not minimize every power error; it is retained because it improves joint detection and makes the activity-sensitive results more consistent across training runs. The complete model consequently provides the best MAE and macro-F1 rather than the best value for every individual metric.

D. Effects of the Training Strategies

Table VI evaluates focal-event mixing and appliance-wise balancing independently and jointly with the architecture fixed.

Focal-event mixing reduces ON MAE from 357.808 to 301.623 W and increases macro-F1 from 0.605 to 0.649, while overall MAE rises from 21.164 to 21.679 W. This pattern is consistent with the strategy's emphasis on active-appliance exposure rather than uniform improvement across all samples. In contrast, appliance-wise balancing independently improves all four evaluation metrics, reducing MAE from 21.164 to 20.929 W while increasing macro-F1 from 0.605 to 0.661. Combining both strategies yields the lowest validation MAE (20.791 W), lowest SAE (15.016 W), and highest macro-F1 (0.687). However, focal-event mixing alone retains the lowest ON MAE (301.623 W), indicating complementary training effects rather than uniform improvements across all metrics.

Figure 6 examines performance across different residual-background power levels. Focal-event mixing improves kettle and microwave reconstruction in several background strata, while washer recall increases across most evaluated intervals. However, performance deteriorates in some low-background conditions, indicating that the benefits are background-dependent rather than uniform across all operating conditions.

Figure 7 illustrates the numerical effect of appliance-wise loss balancing. Under fixed weighting, the median state-loss contribution represents only 2.0% and 0.8% of the corresponding power loss for kettle and microwave, respectively. Bounded appliance-wise balancing increases these ratios to 28.5% and 15.1%, reducing the numerical disparity between the two objectives. These changes are consistent with the validation improvements in Table VI, although they do not establish equivalent gradient contributions.

E. Limitations and Future Work

The results reveal limitations in appliance-specific improvements, background sensitivity, and cross-house evaluation coverage. Although the complete framework improves REFIT microwave detection relative to the reproduced baselines, the training-strategy ablation does not show a corresponding microwave-specific F1 gain (0.352 ± 0.026 versus 0.348 ± 0.027). Thus, the complete framework's improvements cannot be attributed uniformly to the two training strategies. Earlier neural and multi-appliance studies likewise report weaker or highly variable microwave performance [5], [8], [16], while SAMNet shows that similar-power background loads can be mistaken for the target appliance [14]. Short, intermittent activations and overlapping background loads compound this ambiguity.

Performance also remains sensitive to residual-background demand. Focal-event mixing reduces microwave errors in several background strata (Fig. 6) but does not eliminate them. Under high aggregate demand, overlapping non-target activity can obscure transitions even when the target signature is unchanged. Evaluation is also limited to one held-out test house per dataset. Future work should extend cross-house evaluation and jointly model appliance-signature shifts and changes in household background composition.

VI. CONCLUSION

This paper presented MART-NILM, a compact framework for joint multi-appliance power estimation and multi-label ON-OFF state detection from low-frequency aggregate measurements. The architecture combines shared temporal encoding, appliance-specific refinement, and cross-appliance feature exchange, while two training-only strategies address limited event-background coverage and appliance-specific power-state loss imbalance. On the held-out UK-DALE and REFIT test houses, MART-NILM achieves appliance-average MAEs of 7.3 and 10.9 W and macro-F1 scores of 0.88 and 0.72, respectively. Relative to MAT-Conv, the strongest reproduced baseline in overall performance, ON MAE decreases by 58.4% and 34.4% across the two datasets, while the proposed model uses 54.6% fewer parameters. Architecture ablations highlight the importance of shared temporal modeling and appliance-specific refinement, while training-strategy ablations reveal complementary effects on active-state reconstruction and overall joint prediction. Together, these findings support compact joint power-state estimation as a smart-meter observation layer for residential demand-side monitoring.

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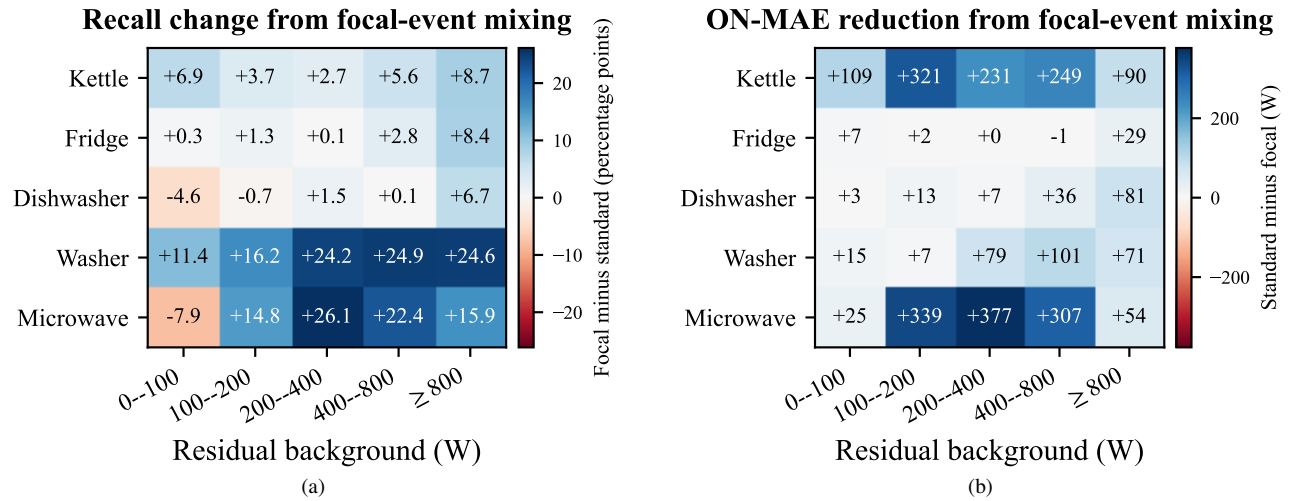


Fig. 6. Effect of focal-event mixing across validation background-demand intervals: (a) active-state recall change in percentage points; (b) ON MAE reduction in watts. Positive values indicate improvement.

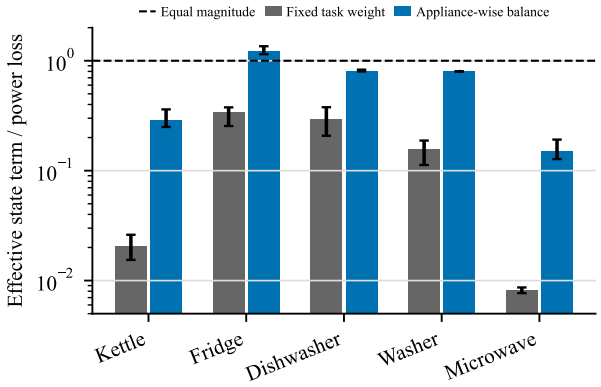


Fig. 7. Power-state scale alignment produced by appliance-wise balancing. Bars show the median effective state-term-to-power-loss ratio pooled over epochs 20 onward and three seeds; error bars denote the interquartile range.

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